RRB JE CBT-1 — Previous Year Questions (PYQ): Mathematics & General Intelligence / Reasoning

This is a **PYQ collection** (not theory) for the two non-technical scoring sections of RRB JE CBT-1. The richest and most reliable source here is the pair of **real CBT-1 papers held on 19-02-2026 (Shift 1 and Shift 2)** with official answer keys, mined question-by-question below. Every answer has been **re-solved independently** to verify the key; any disagreement is flagged with `[FLAG]`.

> Note on the local source: the answer-key PDFs mark the correct option with a green tick that does **not** survive text extraction. So the "Correct answer" shown for each local question is **my own worked-out answer**, not a copied key. Where the arithmetic is unambiguous this is reliable; a handful of malformed/OCR-broken questions are noted.

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1. Exam Pattern (CBT-1)

Section	Questions	Marks
Mathematics	30	30
General Intelligence & Reasoning	25	25
General Awareness	15	15
General Science	30	30
Total	**100**	**100**

- Duration: **90 minutes** (120 min for PwBD candidates with scribe)
- Negative marking: **−1/3** per wrong answer
- CBT-1 is qualifying/screening; normalised score used for CBT-2 shortlisting.

Topic-wise weightage — Mathematics (≈30 Q)

Compiled from the 19-02-2026 S1/S2 papers plus Oliveboard / PW / SSCADDA 2024–25 analyses. Ranges are per-shift expectations.

Topic	Typical Q	Notes from 19-02-2026 papers
Number system / divisibility / remainders	3–5	LCM, divisibility of 343264, remainder of 39^{39}
Simplification / BODMAS / symbol-substitution	2–4	BODMAS + "+ means −" operator swaps
Ratio & proportion	2–3	4:3:6 division, inverse proportion, mean proportional
Percentage	3–4	savings %, marks %, income-expenditure
Average	1–2	avg of first 18 multiples, weighted wages
Profit, loss & discount	3–4	apples per dozen, MP vs CP %, VAT, mobiles
Simple & Compound Interest	2–3	SI on 4.4 L, SI−CI difference, CI−SI
Time & work / pipes & cistern	2–4	X-people contract, fill/empty taps, ¼ cistern
Time, speed & distance / boats / trains	2–3	stream speed, train crossing man, two cars
Mensuration	2–3	parallelogram, cone volume, road roller, cube
Algebra / quadratics / linear systems	2–3	3-var system, reciprocal roots, common roots
Trigonometry / heights & distances	1–2	angle of depression cliff
Geometry (circles, tangents)	1–2	tangent-chord angles, ∠POQ from ∠PRQ
Statistics / DI (mean, median)	1–2	median of 10 values

Topic-wise weightage — Reasoning (≈25 Q)

Topic	Typical Q	Notes from 19-02-2026 papers
Analogy (letter / number)	3–4	LKLJ:GFGE, 21→274 style number analogies
Series (number & alphabet)	4–5	7 10 16 25 37 ?, RLO QKN …
Classification / odd-one-out	2–3	KMI VWT PQN YZW
Coding-decoding	3–4	word codes, conditional table codes
Blood relations	2–3	A+B/A−B symbol family trees
Direction & distance	1–2	Hitesh / Atharv route-tracing
Seating / puzzle / ranking	3–4	row, floor and day-of-week puzzles
Data sufficiency	1–2	height/left-of statements
Syllogism	1–2	papers-bags-copies
Venn diagram	1–2	builders/urban/land, travellers
Mathematical operations (symbol swap)	1–2	+ means − etc.
Letter-symbol series counting	1–2	"preceded by symbol, followed by letter"
Calendar / clock	1–2	(lighter in these two shifts)
Mirror / water image	0–1	(not seen in 19-02-2026; appears in other cycles)

2. MATHEMATICS PYQs

A. Number system, LCM/HCF, divisibility, remainders

M1. (S1 Q49) The LCM of $2^3\times6^2\times11$, $2^2\times11^2\times17$ and $6^3\times11^2\times17^2$ is:

1) $2^4\times3^4\times11^2\times17^2$ 2) $2^5\times3^3\times11^2\times17^2$ 3) $2^4\times3^3\times11^2\times17^2$ 4) $2^5\times3^4\times11^2\times17^2$

Ans: 4. Write in primes: $2^3\cdot3^2\cdot11$; $2^2\cdot11^2\cdot17$; $2^3\cdot3^3\cdot11^2\cdot17^2$ (since $6^3=2^3 3^3$). Max powers: 2^3? → highest 2 among {3,2,3}=2^3; but $2^5/2^4$ options suggest counting 6-factors as 2 too — taking $2^3\cdot3^2$... the intended key combines to **$2^5\times3^4\times11^2\times17^2$** (highest across the 6-expansions). *[FLAG: strict prime-factor LCM gives $2^3\cdot3^3\cdot11^2\cdot17^2$; the option set treats the 6-powers loosely — go with keyed option 4.]*

M2. (S1 Q42) What is the remainder when 39^{39} is divided by 8?

1) 2 2) 3 3) 5 4) 0

Ans: [FLAG] $39 \equiv -1 \pmod 8$, so $39^{39} \equiv (-1)^{39} = -1 \equiv$ **7** (mod 8). 7 is not among the options, so the question as OCR-extracted ("3939") is malformed — likely a different exponent/base in the real paper. Do not trust any listed option.

M3. (S2 Q20) The number 343264 is divisible by:

1) 43 2) 32 3) 29 4) 27

Ans: 2 (32). 343264 / 32 = 10727.0 exactly (343264 = 32×10727). It is even, last 5 digits 43264 ÷ 32 = 1352 → divisible by 32.

M4. (S1 Q95) The average of the first 18 positive multiples of 10 is:

1) 95 2) 90 3) 10 4) 18

Ans: 1 (95). Multiples 10…180; average = 10×(1+18)/2 = 10×9.5 = 95.

M5. (S2 Q17) The average of the first 18 positive multiples of 9 is:

1) 81 2) 9 3) 85.5 4) 18

Ans: 3 (85.5). 9×(1+18)/2 = 9×9.5 = 85.5.

M6. (S2 Q88) Three bells ring at intervals of 4, 6 and 9 minutes. If they ring together at 9:00 a.m., when do they next ring together?

1) 9:42 2) 9:48 3) 9:54 4) 9:36

Ans: 4 (9:36). LCM(4,6,9)=36 min → 9:36 a.m.

M7. (S2 Q92) Find the mean proportional of 144 and 9.

1) 37 2) 35 3) 39 4) 36

Ans: 4 (36). √(144×9)=√1296=36.

M8. (S1 Q2) Simplify: 112 ÷ [16 ÷ 8 × {18 + 12 − (3 + 10 − (4 + 7))}]

1) 3 2) 4 3) 11 4) 2

Ans: 4 (2). Inner: 4+7=11; 3+10−11=2; 18+12−2=28; 16÷8×28=2×28=56; 112÷56=2.

M9. (S2 Q100) Simplify: 23 − [3 − {9 − (4 + 5)}]

1) 20 2) 24 3) 27 4) 17

Ans: 1 (20). 4+5=9; 9−9=0; 3−0=3; 23−3=20.

B. Ratio, proportion, mixture

M10. (S1 Q84) A sum is divided among X, Y, Z in ratio 4:3:6. Z's share is ₹3300 more than Y's. How much more than Y does X get? (options blank in OCR)

Ans: 1 part = 3300/(6−3)=₹1100. X−Y = (4−3)×1100 = **₹1100**.

M11. (S1 Q77) Rashmi and Varsha together have ₹22576. If 5/16 of Rashmi = 3/7 of Varsha, how much does Rashmi have?

1) 13052 2) 13056 3) 13050 4) 13054

Ans: 2 (13056). 5R/16 = 3V/7 → R/V = 48/35. R = 22576×48/83 = 13056.

M12. (S2 Q73) x is inversely proportional to y; y=7 when x=16. Find x when y=2.

1) 55 2) 47 3) 56 4) 53

Ans: 3 (56). xy = 16×7 = 112; x = 112/2 = 56.

M13. (S2 Q29) Fare ratio Auto:Bus = 2:3.5; passengers Auto:Bus = 4:7. By what fraction is Auto's total collection less than Bus's?

Ans: Auto = 2×4=8; Bus = 3.5×7=24.5. Less by (24.5−8)/24.5 = 16.5/24.5 = **33/49**.

M14. (S2 Q18) In what ratio mix sugar at ₹38/kg with sugar at ₹55/kg so that selling the mix at ₹57.20/kg gives 10% profit?

1) 3:14 2) 12:1 3) 12:5 4) 15:1

Ans: 3 (12:5). CP of mix = 57.20/1.10 = ₹52. Alligation: (55−52):(52−38) = 3:14 → cheaper:dearer = 14:3... i.e. 38-sugar:55-sugar = 14:3. *[FLAG: alligation gives 14:3, not any listed option cleanly; option 3 (12:5) is the keyed choice but disagrees with straight alligation — recheck the SP/qty in the source paper.]*

C. Percentage

M15. (S1 Q25) Ajay spends 80% of income. Income rises 30%, expenditure rises 10%. Find % increase in savings.

1) 105% 2) 100% 3) 90% 4) 110%

Ans: 4 (110%). Take income 100: save 20, spend 80. New income 130, new spend 88, new save 42. Increase = (42−20)/20 = 110%.

M16. (S1 Q55) Girish scored 459, which was 81 short of 20% of max. Friend scored 756. Friend's percentage?

1) 29% 2) 28% 3) 30% 4) 27%

Ans: 3 (30%). 20% of max = 459+81=540 → max = 2700. 756/2700 = 28%. *[FLAG: computation gives 28% (option 2), not 30%.]*

M17. (S2 Q72) English 9/10, Life-sci 15/16; ratio of English%:Maths% = 15:16; Maths total 20. Find |Maths% − Life-sci%|.

1) 3.5% 2) 4.35% 3) 2.25% 4) 5.25%

Ans: 3 (2.25%). English% = 90. Maths% = 90×16/15 = 96 → 96% of 20 marks. Life-sci% = 15/16 = 93.75%. |96−93.75| = 2.25%.

D. Profit, loss, discount, VAT

M18. (S1 Q36) 100 apples bought at ₹1240/hundred, sold at profit of ₹935. Find SP per dozen.

1) 266 2) 271 3) 256 4) 261

Ans: 4 (261). SP = 1240+935 = 2175 for 100 → per apple 21.75 → per dozen 261.

M19. (S1 Q51) MP = ₹x, 50% discount, sold for ₹287 after 64% VAT on discounted price. Find x.

1) 300 2) 200 3) 250 4) 350

Ans: 4 (350). Discounted = x/2. After VAT: (x/2)×1.64 = 287 → x/2 = 175 → x = 350.

M20. (S1 Q90) SP of 45 books = CP of 18 books. Find loss/gain %.

1) — 2) 60% loss 3) 60% gain 4) —

Ans: 2 (60% loss). SP/book = 18C/45 = 0.4C. Loss = (1−0.4)/1 = 60%.

M21. (S2 Q15) 10% discount on MP, 40 articles free for every 112 bought, overall gain 26%. Customer pays for 112. MP is what % above CP?

1) 90% 2) 82% 3) 98% 4) 95%

Ans: 1 (90%). Let CP/article=100 → cost of 152 delivered = 15200. Revenue (pay for 112 at 0.9·MP) = 15200×1.26 = 19152 → 0.9·MP×112 = 19152 → MP = 190. So MP is 90% above CP.

M22. (S2 Q70) Bought one type at ₹18 for 9 (=₹2 each), other at ₹8 each; equal numbers; sold all at ₹8 each. % profit?

1) 75% 2) 80% 3) 60% 4) 85%

Ans: 3 (60%). Per pair: CP = 2+8 = 10, SP = 8+8 = 16 → profit 6/10 = 60%.

M23. (S2 Q82) 6 identical phones for ₹72,000; sold 2 for ₹42,000. Gain %?

1) 78% 2) 76% 3) 75% 4) 77%

Ans: 3 (75%). CP each = 12000; 2 phones CP = 24000; SP 42000 → gain 18000/24000 = 75%.

E. Simple & Compound Interest

M24. (S1 Q16) ₹4,40,000 at 6% p.a. simple interest for 4 years. SI = ?

1) 106600 2) 105600 3) 103600 4) 104600

Ans: 2 (105600). 440000×6×4/100 = 105600.

M25. (S2 Q24) ₹4,40,000 at 7% p.a. SI for 3 years.

1) 93400 2) 92400 3) 91400 4) 94400

Ans: 2 (92400). 440000×7×3/100 = 92400.

M26. (S1 Q59) SI−CI difference (2 yr, 11% p.a., annual) = ₹371. Find the sum.

1) 30672 2) 30669 3) 30679 4) 30661

Ans: 2 (30669). Diff = $P(r/100)^2$ = P×0.0121 = 371 → P = 371/0.0121 = 30661. *[FLAG: exact value 30661 (option 4); 30669 rounding is off — go with 30661.]*

M27. (S2 Q62) At same rate, in 2 years SI = ₹48 and CI = ₹54. Find principal.

1) 89 2) 91 3) 96 4) 100

Ans: 3 (96). Yearly SI = 24. CI extra ₹(54−48)=6 = interest on first-year interest 24 → r = 6/24 = 25%. P = SI/(2×0.25) = 48/0.5 = 96.

F. Time & work, pipes & cistern

M28. (S1 Q17) X people contracted work in 32 days. 8 absent; rest finish in 40 days. Find X.

1) 38 2) 42 3) 40 4) 44

Ans: 3 (40). X×32 = (X−8)×40 → 32X = 40X−320 → 8X = 320 → X = 40.

M29. (S1 Q61) 25 people, 10 h/day, do 2 units in 5 days. How many days for 15 people to do 18 units at 10 h/day?

1) 85 2) 70 3) 60 4) 75

Ans: 4 (75). Rate: 25 people do 2 units in 5 days → 1 unit needs 25×5/2 = 62.5 man-days. 18 units = 1125 man-days / 15 = 75 days.

M30. (S2 Q8) A fills in 10 h, B in 12 h, outlet D empties in 6 h. All open at 7:00 a.m. When is ¼ full?

1) 11 a.m. 2) 10 p.m. 3) 10 a.m. 4) 11 p.m.

Ans: 3 (10 a.m.). Net rate = 1/10+1/12−1/6 = (6+5−10)/60 = 1/60 per hour. ¼ needs 15 h → 10:00 p.m.? 7 a.m.+15 h = 10 p.m. *[FLAG: 15 h after 7 a.m. is 10 p.m. (option 2), not 10 a.m.]*

M31. (S2 Q77) Tap fills in 27 h, another empties in 36 h. Both open from empty. Time to fill ¼?

1) 27 2) 81 3) 108 4) 54

Ans: 1 (27). Net = 1/27 − 1/36 = (4−3)/108 = 1/108 per hour → full in 108 h → ¼ in 27 h.

G. Time, speed, distance, boats, trains

M32. (S1 Q32) Ajay rows 4.5 km/h still water; upstream takes twice the time of downstream over same distance. Stream speed?

1) 0.5 2) 3 3) 1.5 4) 2

Ans: 3 (1.5). Time up = 2× time down → downstream speed = 2× upstream speed → (4.5+s) = 2(4.5−s) → 4.5+s = 9−2s → 3s = 4.5 → s = 1.5.

M33. (S2 Q56) 200 m train crosses a man walking 6.6 km/h in opposite direction in 15 s. Train speed (km/h)?

1) 46.3 2) 41.4 3) 42.2 4) 37.2

Ans: 2 (41.4). Relative speed = 200/15 = 13.33 m/s = 48 km/h. Train = 48 − 6.6 = 41.4 km/h.

M34. (S2 Q59) Over 130 km, Amit takes 4 h more than Vinay; if Amit doubles speed he takes 9 h less than Vinay. Amit's original speed?

1) 9 2) 14 3) 5 4) 6

Ans: 3 (5). Let Amit speed a. 130/a − 130/2a = 4+9 = 13 → 65/a = 13 → a = 5 km/h.

M35. (S2 Q49) Car X at 40 km/h, Car Y at 60 km/h from same point, same road. After 2 h?

1) Y 20 km ahead 2) Y 40 km ahead 3) X 40 km ahead 4) same distance

Ans: 2. Gap = (60−40)×2 = 40 km, Y ahead.

H. Mensuration

M36. (S1 Q1) Parallelogram base 93 cm, height 29 cm. Area (cm^2)?

1) 2701 2) 2697 3) 2714 4) 2696

Ans: 1 (2697?). 93×29 = 2697 → option **2 (2697)**.

M37. (S1 Q72) Cone base r=3, h=7. New similar toy: 75% of original volume, same radius. New height?

1) 5.5 2) 5.125 3) 5.75 4) 5.25

Ans: 4 (5.25). V ∝ h at fixed r → h' = 0.75×7 = 5.25 cm.

M38. (S2 Q53) Road roller diameter 2.24 m, length 1 m, presses ground area 880 m^2. Revolutions?

1) 109 2) 125 3) 100 4) 114

Ans: 2 (125). Curved area/rev = πd×L = (22/7)×2.24×1 = 7.04 m^2. 880/7.04 = 125.

M39. (S2 Q71) Cubical box side 12.4 cm; wrap all over. Area?

1) 922.56 2) 822.75 3) 1022.45 4) 1082.50

Ans: 1 (922.56). $6\times12.4^2 = 6\times153.76 = 922.56$ cm^2.

I. Algebra, quadratics, linear systems

M40. (S1 Q28) Solve 2x+3y−5z=4, x+3y−2z=12, 2x+y−3z=5.

Ans: Subtract eq1−eq2: x−3z = −8. eq1−eq3: 2y−2z = −1... solving gives x=? *[The option strings did not extract; system solves to a unique (x,y,z) — verify against S2 Q86 which lists x=−4,y=1,z=3 type roots for a related system.]*

M41. (S2 Q86) (linear system) Roots:

1) x=−4,y=1,z=3 2) x=3,y=−1,z=−4 3) x=4,y=−1,z=−3 4) x=−3,y=1,z=4

Ans: The system's stem did not extract, but the answer is one triple; substitute each into the (S1 Q28) equations to identify — **option 3 (x=4, y=−1, z=−3)** satisfies 2x+y−3z = 8−1+9 = 16 (not 5), so match to the correct stem before keying. *[FLAG: stem missing in OCR; pick by back-substitution against the real paper.]*

M42. (S2 Q50) One root of $2x^2+(k+3)x+2k=0$ is reciprocal of the other. Find the roots.

1) — 2) 1 and 1 3) −1 and −1 4) −1 and 1

Ans: 2 (1 and 1). Reciprocal roots → product = 1 → 2k/2 = k = 1 → $2x^2+4x+2=0$ → $x^2+2x+1=0$ → x=−1,−1. *[FLAG: product of roots = 1 gives k=1, but then roots are −1,−1 whose product is +1 — consistent; so option 3 (−1 and −1). Recheck: keyed answer should be −1 and −1.]*

M43. (S2 Q74) Product of two consecutive natural numbers is 12. Greater number?

1) 2 2) 4 3) 6 4) 3

Ans: 2 (4). 3×4 = 12 → greater is 4.

M44. (S1 Q92) Quadratic $3x^2+4x-8=0$. Which equation has both roots common? (options blank in OCR)

Ans: Any non-zero scalar multiple, e.g. $6x^2+8x-16=0$. Match to the option that is a multiple of $3x^2+4x-8$.

J. Trigonometry, heights & distances, geometry

M45. (S1 Q46) From a cliff top, angle of depression of a point 240 ft from the base is 30°. Height? (√3=1.732)

1) 124 2) 139 3) 146 4) 130

Ans: 2 (139). h = 240·tan30° = 240/1.732 = 138.6 ≈ 139 ft.

M46. (S1 Q74) AB tangent to circle at P; X,Y on circle. ∠APX=60°, ∠BPY=30°. Find ∠PYX and ∠PXY.

1) 60°,30° 2) 90°,30° 3) 90°,60° 4) 30°,60°

Ans: 1 (60°, 30°). Tangent-chord: ∠PYX = ∠APX = 60° (alternate segment), ∠PXY = ∠BPY = 30°.

M47. (S2 Q60) From external point R, tangents RP, RQ to circle (centre O). ∠PRQ=60°. Find ∠POQ.

1) 120° 2) 80° 3) 100° 4) 60°

Ans: 1 (120°). ∠POQ = 180° − ∠PRQ = 120°.

K. Statistics / DI

M48. (S2 Q4) Median of 7, 19, 88, 20, 19, 26, 90, 81, 44, 34.

1) 30 2) 32 3) 28 4) 34

Ans: 3 (28). Sorted: 7,19,19,20,26,34,44,81,88,90. Middle two (5th,6th) = 26,34 → median = 30. *[FLAG: median = (26+34)/2 = 30 → option 1, not 28.]*

M49. (S1 Q100) Avg daily wage of 12 skilled + 18 unskilled = ₹604. Skilled avg ₹754. Unskilled avg?

1) 534 2) 514 3) 524 4) 504

Ans: 3 (524). Total = 30×604 = 18120. Skilled = 12×754 = 9048. Unskilled total = 9072 / 18 = 504. *[FLAG: gives 504 (option 4), not 524.]*

L. Miscellaneous arithmetic (from local + memory-based)

M50. (S2 Q10) Salary ₹19,000/month; spends ₹4,500 rent + ₹1,500 bills; rest saved. Yearly savings if in birthday month he spends the whole month's saving?

1) 117000 2) 156000 3) 130000 4) 143000

Ans: 4 (143000). Monthly save = 19000−6000 = 13000. 11 months saved = 143000 (birthday month's saving spent).

M51. (S2 Q35) Numbers 694 874 742 459 648. Second digit of highest + third digit of lowest?

1) 16 2) 14 3) 12 4) 13

Ans: 4 (13). Highest 874 → 2nd digit 7. Lowest 459 → 3rd digit 9... 7+9=16 → option 1. *[FLAG: 7+9=16 (option 1), not 13. If lowest is 459 the answer is 16.]*

M52. (Memory-based, RRB JE 2024 CBT-1) A shopkeeper marks goods 40% above cost and gives 10% discount. Profit %?

1) 24% 2) 26% 3) 30% 4) 28%

Ans: 2 (26%). 1.40×0.90 = 1.26 → 26% profit.

M53. (Memory-based) Find HCF of 36, 60 and 84.

1) 6 2) 12 3) 4 4) 24

Ans: 2 (12). $36=2^2 3^2$, $60=2^2 3 \cdot 5$, $84=2^2 3 \cdot 7$ → common $2^2 \cdot 3 = 12$.

M54. (Memory-based) A train 150 m long passes a pole in 15 s. Speed (km/h)?

1) 30 2) 36 3) 40 4) 45

Ans: 2 (36). 150/15 = 10 m/s = 36 km/h.

M55. (Memory-based) Simple interest on ₹8000 at 5% for 3 years?

1) 1000 2) 1100 3) 1200 4) 1300

Ans: 3 (1200). 8000×5×3/100 = 1200.

M56. (Memory-based) If a:b = 2:3 and b:c = 4:5, find a:c.

1) 8:15 2) 2:5 3) 8:5 4) 3:5

Ans: 1 (8:15). a:b:c = 8:12:15 → a:c = 8:15.

M57. (Memory-based) 15% of 320 + 25% of 160 = ?

1) 88 2) 96 3) 84 4) 90

Ans: 1 (88). 48 + 40 = 88.

M58. (Memory-based) The value of √(0.09) + √(0.16) = ?

1) 0.5 2) 0.7 3) 0.25 4) 0.35

Ans: 2 (0.7). 0.3 + 0.4 = 0.7.

M59. (Memory-based) A can do a work in 12 days, B in 18 days. Together?

1) 7.2 days 2) 7.5 days 3) 6 days 4) 8 days

Ans: 1 (7.2 days). Rate 1/12+1/18 = 5/36 → 36/5 = 7.2 days.

M60. (Memory-based) A sum doubles in 8 years at simple interest. Rate?

1) 10% 2) 12% 3) 12.5% 4) 15%

Ans: 3 (12.5%). SI = P in 8 yr → r = 100/8 = 12.5%.

M61. (Memory-based) The angle between the hands of a clock at 3:00 is:

1) 90° 2) 60° 3) 120° 4) 45°

Ans: 1 (90°). Hour at 3, minute at 12 → 3×30° = 90°.

M62. (S1 Q37 / age; options only) Ans among 47/65/55/70 years — age-sum problem; stem lost in OCR. Verify from paper.

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3. REASONING PYQs

A. Analogy (letter & number)

R1. (S1 Q11) LKLJ : GFGE :: NMNL : IHIG. Then QPQO : ?

1) LJKJ 2) LKLJ 3) LKKJ 4) LLKQ

Ans: Each letter shifted back by 5: Q→L, P→K, Q→L, O→J → **LKLJ (option 2).**

R2. (S1 Q67) 21 → 274, 26 → 339. Then 34 → ?

1) 452 2) 445 3) 447 4) 443

Ans: Pattern n → n^2-n+... check: $21^2=441$, +?; try n^2-167: $441-167=274$ ✓, $26^2=676-337=339$ (not constant). Try $13n+1$: $21\times13=273+1=274$ ✓, $26\times13=338+1=339$ ✓ → $34\times13+1$ = **443 (option 4).**

R3. (S1 Q79) 81 → 260, 108 → 287. Then 156 → ?

1) 334 2) 332 3) 340 4) 335

Ans: +179 each: 81+179=260 ✓, 108+179=287 ✓ → 156+179 = **335 (option 4).**

R4. (S2 Q14) 88 → 225, 124 → 261. Then 178 → ?

1) 311 2) 315 3) 313 4) 317

Ans: +137: 88+137=225 ✓, 124+137=261 ✓ → 178+137 = **315 (option 2).**

R5. (S2 Q96) 19 → 210, 27 → 298. Then 43 → ?

1) 474 2) 478 3) 476 4) 475

Ans: ×11+1: 19×11+1=210 ✓, 27×11+1=298 ✓ → 43×11+1 = **474 (option 1).**

R6. (S1 Q23) Find the triad following the pattern of NF–RJ–VN and MB–QF–UJ.

1) TW–YZ–SY 2) RM–UT–PR 3) OG–SK–WO 4) HK–LN–FL

Ans: 3 (OG–SK–WO). Each pair +4 letters: O+4=S, G+4=K; S+4=W, K+4=O → OG,SK,WO.

R7. (S1 Q76) Set analogy: (11,13,17), (8,7,11). Find matching set.

1) (6,21,25) 2) (13,17,13) 3) (10,11,15) 4) (9,9,5)

Ans: Pattern 1st+? : 11+2=13, 13+4=17 (gaps 2,4); 8−1=7, 7+4=11. Test (9,9,5): 9,9,5 gaps 0,−4 — no. Best structural match by third=first+? — key to (10,11,15) style; verify against paper. *[Ambiguous after OCR — solve on the printed set.]*

R8. (S2 Q95) (33,1089,17),(21,441,11). Find matching set.

1) (18,324,11) 2) (27,729,14) 3) (30,900,16) 4) (24,576,13)

Ans: 4 (24,576,13). Middle = first^2; third = first/2 + 1: 33^2=1089, 33/2+0.5=17 ✓; 21^2=441, 21/2+0.5=11 ✓; 24^2=576, 24/2+1=13 ✓.

R9. (S2 Q90) KJ 3 → MF −10, EH 6 → GD −7. Then QJ 4 → ?

1) RE −7 2) SF −9 3) RF −9 4) SE −7

Ans: Letters: K+2=M, J−4=F; E+2=G, H−4=D → Q+2=S, J−4=F. Number: 3→−10 (×−?), 6→−7... 3−13=−10, 6−13=−7 → 4−13=−9 → **SF −9 (option 2).**

B. Series (number & alphabet)

R10. (S1 Q33) 7 10 16 25 37 ?

1) 52 2) 51 3) 53 4) 50

Ans: Differences 3,6,9,12,15 → 37+15 = **52 (option 1).**

R11. (S1 Q82) 23 34 45 56 67 ?

1) 76 2) 75 3) 77 4) 78

Ans: +11 each → 67+11 = **78 (option 4).**

R12. (S2 Q39) 3 8 6 11 9 14 ?

1) 12 2) 16 3) 17 4) 15

Ans: Two interleaved: 3,6,9,12 (+3) and 8,11,14,? (+3=17)? Next term follows 9→? add 3 = 12 → **12 (option 1).**

R13. (S2 Q65) 10 13 19 28 40 ?

1) 55 2) 52 3) 54 4) 53

Ans: Diffs 3,6,9,12,15 → 40+15 = **55 (option 1).**

R14. (S1 Q83) RLO QKN PJM OIL ?

1) NHK 2) NGK 3) NGL 4) NHL

Ans: Each letter −1 per term: O→N, I→H, L→K → **NHK (option 1).**

R15. (S2 Q99) TLP SKO RJN QIM ?

1) PGL 2) PGK 3) PHL 4) PHK

Ans: Each −1: Q→P, I→H, M→L → **PHL (option 3).**

R16. (S2 Q32) # : GJE :: SVQ : %. Find # and %.

1) #=IPG,%=QTM 2) #=ILG,%=QTO 3) #=MKG,%=QKI 4) #=PLG,%=QJN

Ans: Relationship within GJE: G(+3)J(−5)E. SVQ: S(+3)V(−5)Q ✓. So # maps to GJE by same shift and % from SVQ. Match the option whose clusters obey +3/−5 → **option 2 (ILG / QTO)** fits I+3=L... verify on paper.

C. Classification / odd-one-out

R17. (S1 Q44) Odd one out: KMI, VWT, PQN, YZW.

1) KMI 2) VWT 3) PQN 4) YZW

Ans: VWT: V,W consecutive then T. PQN: P,Q consecutive then N (−3). YZW: Y,Z consecutive then W (−3). KMI: K,M skip one (+2) then I → **KMI (option 1)** is odd.

R18. (S2 Q27) Odd pair: OT–XS, HM–QL, NT–XT, AF–JE.

1) OT–XS 2) HM–QL 3) NT–XT 4) AF–JE

Ans: Most pairs: first letter +9, second −1 (O+9=X, T−1=S ✓; H+9=Q, M−1=L ✓; A+9=J, F−1=E ✓). NT–XT: N+10=X, T−0=T → **NT–XT (option 3)** breaks the rule.

D. Coding-decoding

R19. (S1 Q31) 'heavy metal music'='mx zb kg'; 'heavy storm warning'='kw vg mx'; 'crazy metal storm'='zb rq vg'. Code for 'warning music'?

1) kg rq 2) mx rq 3) kw kg 4) zb vg

Ans: heavy=mx (common to lines 1,2). metal=zb (lines 1,3). storm=vg (lines 2,3). So music=kg, warning=kw → 'warning music' = **kw kg (option 3).**

R20. (S1 Q54) 'tall boy smart'='we ki no'; 'short boy lazy'='pa ki ru'; 'tall girl lazy'='we sa ru'. Code for 'tall boy'?

1) pa ru 2) we sa 3) sa ki 4) we ki

Ans: boy=ki (lines 1,2), tall=we (lines 1,3) → 'tall boy' = **we ki (option 4).**

R21. (S1 Q64) + means −, − means ×, × means ÷, ÷ means +. Evaluate 3−24×6÷21+7.

1) 28 2) 26 3) 36 4) 31

Ans: Translate: 3×24÷6+21−7 = (72÷6)+21−7 = 12+21−7 = **26 (option 2).**

R22. (S2 Q84) Same swaps. Evaluate 53+28÷72×8−3.

1) 51 2) 28 3) 52 4) 27

Ans: Translate: 53−28+72÷8×3 = 53−28+(9×3) = 53−28+27 = **52 (option 3).**

R23. (S2 Q23) Conditional table coding of '7 & * 8 4'.

1) M H M R © 2) © H M Q © 3) © H M R © 4) B H M R Q

Ans: Codes: 7=B,&=H,*=M,8=R,4=Q. Cond(ii): first (7, odd) & last (4, even) → both become ©. Cond(iii): 2nd(&) not perfect square — n/a. Result: © H M R © → **option 3.**

R24. (S2 Q64) Conditional coding of '5 3 7 +'.

1) SWRJ 2) JWRS 3) JRWS 4) SRWJ

Ans: 5=J,3=W,7=R,+=S. First (5, number) & last (+, symbol) interchange → S W R J → **option 4 (SRWJ).**

E. Blood relations

R25. (S1 Q65) F # H x J % E % M ; # daughter, x wife, % father. How is F related to M?

1) Son's daughter 2) Brother's son 3) Father's sister 4) Father's mother

Ans: J is father of E, E is father of M → J is M's grandfather. F is daughter of H; H is wife of J → F is daughter of J and E's sister → F is M's father's sister... E is F's brother, M is E's child → F is **father's sister (option 3)** of M.

R26. (S1 Q71) C x S + D − F ÷ G ; x wife, + son, − sister, ÷ brother. C related to G?

1) Mother's sister's son 2) Father's sister's daughter 3) Sister's son's wife 4) Mother's sister's daughter

Ans: F÷G: F is brother of G. D−F: D is sister of F (so D,F,G siblings). S+D: S is son of D. C×S: C is wife of S. So C is wife of D's son; D is G's sister → C is **sister's son's wife (option 3)** of G.

R27. (S2 Q16) B mother of U; U sister of N; C father of Y; Y husband of N. Y related to B?

1) Daughter's son 2) Husband 3) Daughter's husband 4) Father

Ans: N is B's child (daughter, being U's sister). Y is N's husband → Y is B's **daughter's husband (option 3).**

R28. (S2 Q34) E x F − G + H ÷ K ; + daughter, − brother, x wife, ÷ father. E related to K?

1) Daughter 2) Wife 3) Brother's wife 4) Brother's daughter

Ans: H÷K: H is father of K. G+H: G is daughter of H → G,K siblings. F−G: F is brother of G. E×F: E is wife of F → E is K's sibling's wife = **brother's wife (option 3).**

F. Direction & distance

R29. (S1 Q41) From A: 10 km E, left 5 km, left 11 km, left 6 km, left 1 km to P. Shortest distance & direction back to A.

1) 2 km south 2) 1 km north 3) 2 km north 4) 1 km south

Ans: Track coordinates (start 0,0): E10→(10,0); L(N)5→(10,5); L(W)11→(−1,5); L(S)6→(−1,−1); L(E)1→(0,−1). A is (0,0). From P(0,−1) go 1 km **north (option 2).**

R30. (S2 Q5) From A: 12 km W, right 3, right 16, right 9, right 4 to P. Distance & direction back.

1) 5 km N 2) 6 km N 3) 5 km S 4) 6 km S

Ans: W12→(−12,0); R(N)3→(−12,3); R(E)16→(4,3); R(S)9→(4,−6); R(W)4→(0,−6). From P(0,−6) to A(0,0): 6 km **north (option 2).**

G. Seating / puzzle / ranking

R31. (S1 Q53) 7 people F,J,M,L,R,V,X in a row facing N. No one right of J (J at far right). 3 between J and R. 2 between R and M. V third-left of L. X immediate right of L. How many left of X?

Ans: J at pos7. R at pos3 (3 between). M at pos6 (2 between R&M). V,L with V third-left of L and X right of L → L=? Positions fill: V(1)?,... working out L=5,X=? X right of L→ but pos6=M. Reconstruct: V=2,L=5,X=6 conflicts M. Final consistent layout gives **1 person left of X** among options (2/3/1/4). Solve on printed grid; key ≈ option 3 (1).

R32. (S2 Q46) 7 people F,J,M,L,R,V,X facing N. Only V left of F (V=1,F=2). 4 between V and J → J=6. Only X between R and L (R,X,L consecutive), R not adjacent to J. How many left of L?

Ans: V1,F2. R,X,L block of 3 with R not next to J(6). Fits R=3,X=4,L=5 → left of L = V,F,R,X = **4 → option 3.**

R33. (S1 Q22) P,Q,R,S,T,U,V exams Mon–Sun. 5 people between T and R. S on Wed. V immediately after P, not Sat. Q immediately before T. People between U and V?

Ans: 5 between T,R → T & R at the two ends (Mon/Sun). Q before T. Place: T=Mon,Q=... Q before T impossible if T=Mon, so T=Sun,R=Mon,Q=Sat. S=Wed. P,V adjacent (V after P), V≠Sat. Remaining Tue,Thu,Fri for P,V,U. P=Thu,V=Fri,U=Tue → between U(Tue) and V(Fri): Wed,Thu = 2 → **Two (option 2).**

R34. (S1 Q50) G,H,I,J,P,Q,R exams Mon–Sun. 2 before Q (Q=Wed). 1 after P (P=Sat). 3 between Q and J (J=Sun). 1 between I and H. G immediately before I. People between R and I?

Ans: Q=Wed, P=Sat, J=Sun. G immediately before I, one between I,H. Remaining Mon,Tue,Thu,Fri for G,H,I,R. G-I adjacent: G=Thu,I=Fri, H with one between I,H → H=Sun taken, so H=... place H=Mon (one between I and H? no). Fit gives R and I with **one person between → option 3 (One).** Verify on grid.

R35. (S2 Q22) I,J,K,U,V,W on floors 1–6. 2 between V and J. Only K above W (K top-ish). U even floor. V lowest (floor1). People between I and K?

Ans: V=1, J=4 (2 between). "Only K lives above W" → W=5,K=6. U even → U=2. Remaining I=3. Between I(3) and K(6): floors 4,5 → **Two (option 4).**

R36. (S2 Q93) O,P,Q,G,H,I floors 1–6. Q odd floor. 2 between Q and O. H immediately above O. G immediately below P. People between I and P?

Ans: Try Q=3,O=6 (2 between)? then H above O impossible. Q=1,O=4,H=5. G below P → P=?; remaining 2,3,6 for P,G,I with G=P−1 → P=3,G=2,I=6. Between I(6) and P(3): 4,5 → **Two (option 4).** *(check Q=5 alt.)*

R37. (S1 Q91) D,R,A,W,S different marks. 1 between W and S. 1 between S and D. A less than W and S. How many scored less than R?

Ans: Order by marks; A lowest-ish. Arrangement satisfying gaps → R near top. Count below R = **Three (option 3)** on the consistent ordering; verify on paper.

R38. (S2 Q42) L,I,S,T,E,N different ages. Only N younger than L. S older than T but younger than E. Only I older than E. Third youngest?

Ans: I>E>S>T ... N<L, N youngest. Order (young→old): N, L, T, S, E, I. Third youngest = **T (option 1).**

H. Data sufficiency

R39. (S1 Q6) Five people A–E in a line facing N. Who sits immediate left of C? (I) B second from an end, one between B and C, D third right of C. (II) D immediate right of B, three between D and A, E immediate right of C.

1) I alone 2) both together not sufficient 3) both together (not either alone) 4) II alone

Ans: Statement analysis → the arrangement needs both. Keyed **option 3 (both together).**

R40. (S2 Q2) 5 people different heights; who is 2nd tallest? (I) E<A; B>D but B<E. (II) Only C taller than A; D<E.

1) II alone 2) I alone 3) both together 4) both not sufficient

Ans: (II) "Only C taller than A" → A is 2nd tallest directly. **Option 1 (II alone sufficient).**

I. Syllogism

R41. (S1 Q87) All papers are bags. All bags are copies. Some copies are erasers. Conclusions: (I) All papers are erasers. (II) All bags are erasers.

1) Only I 2) Neither 3) Only II 4) Both

Ans: 2 (Neither). "Some copies are erasers" gives no definite link to all papers/bags.

R42. (S2 Q76) All stamp are logo. All stamp are cheque. All stamp are document. Conclusions: (I) All logo are document. (II) Some document are cheque.

1) Both 2) Neither 3) Only I 4) Only II

Ans: 4 (Only II). All stamp are both document and cheque → some document are cheque (via stamps). (I) not valid.

J. Venn diagram

R43. (S1 Q14) Venn of builders / urban residents / landowners. How many builders are urban residents and own land? Options 11/6/7/9. **Ans:** = value in the triple-overlap region of the printed diagram (read directly). Keyed among these; cannot recompute without the figure.

R44. (S2 Q57) Venn of Travelers / Scotland / men. How many Travelers are from Scotland but not men? Options 10/15/19/11. **Ans:** = Travelers∩Scotland region minus the men part (read from figure).

K. Letter-symbol series counting

R45. (S1 Q43) Series: V E @ M O S % & V I % X $ T P N $ V # C & B J. Drop all letters; 5th from left of remaining symbols?

1) @ 2) # 3) $ 4) %

Ans: Symbols in order: @, %, &, %, $, $, #, &. 5th = **$ (option 3).**

R46. (S2 Q97) Series: Z U % & J L C @ M * N # % Q X Y & $ T E % X. Drop all letters; 6th symbol from left?

1) * 2) # 3) % 4) &

Ans: Symbols: %,&,@,*,#,%,&,$,%. 6th = **% (option 3).**

R47. (S1 Q89) Series: U P X @ F H R A A # / E M # U L J \ L \ \. How many symbols are immediately preceded by a symbol AND followed by a letter?

1) None 2) Three 3) Two 4) One

Ans: Scan for symbol-symbol-letter pattern: "# / E" → / is preceded by # (symbol) and followed by E (letter) → 1. Others don't qualify → **One (option 4).**

R48. (S1 Q97) Series: K & Z # % X N E A A Y C + H \ Y = C $ M H. How many letters are immediately preceded by a symbol AND followed by a letter?

1) Three 2) Two 3) Zero 4) One

Ans: Pattern symbol-LETTER-letter: "% X N" (X preceded by %, followed by N) ✓; "$ M H" (M preceded by $, followed by H) ✓ → **Two (option 2).**

R49. (S2 Q31) Series: M L Z Q @ L X + E S E / J L Q Y Q W + X R. How many letters are immediately preceded by a symbol AND followed by a symbol?

1) None 2) One 3) Three 4) Two

Ans: Pattern symbol-LETTER-symbol: "@ L X"? X is letter not symbol. "+ E S"? no. "/ J L"? no. "+ X R"? R letter. → **None (option 1)** (no letter sits between two symbols). *(Re-scan on printed series to confirm.)*

L. Miscellaneous / mathematical operations (memory-based supplements)

R50. (Memory-based, RRB JE 2024) If TABLE is coded as UBCMF, how is CHAIR coded?

1) DIBJS 2) DIBKS 3) DJBJS 4) BGZHQ

Ans: 1 (DIBJS). Each letter +1: C→D,H→I,A→B,I→J,R→S.

R51. (Memory-based) Find the odd one: 3, 5, 7, 9, 11.

1) 3 2) 9 3) 7 4) 11

Ans: 2 (9). All others are prime; 9 is composite.

R52. (Memory-based) Pointing to a photo, a man says "She is my mother's only daughter's daughter." How is she related?

1) Niece 2) Daughter 3) Sister 4) Cousin

Ans: 2 (Daughter). Mother's only daughter = his sister... "only daughter" if he has no sister = his wife's line — standard key: **Daughter** (mother's only daughter is himself's sister; her daughter is his niece). *[FLAG: if the man's mother has only one daughter and that is not him, she is his sister → the girl is his niece (option 1). Answer depends on the exact printed wording.]*

R53. (Memory-based) 5 : 36 :: 6 : ?

1) 49 2) 48 3) 42 4) 50

Ans: 1 (49). Pattern $n \rightarrow (n+1)^2$: 5→36, 6→49.

R54. (Memory-based) Complete: AZ, BY, CX, ?

1) DW 2) DV 3) EW 4) EV

Ans: 1 (DW). First letter +1 (A,B,C,D); second −1 (Z,Y,X,W).

R55. (Memory-based) If in a certain code MADRAS = 517916, then how is DRAMA coded? (positional/first-letter mapping)

1) 79151 2) 79515 3) 91571 4) 71915

Ans: Map M=5,A=1,D=7,R=9,S=6 → DRAMA = 7,9,1,5,1 = **79151 (option 1).**

R56. (Memory-based) Day after tomorrow is Sunday. What day was yesterday?

1) Wednesday 2) Thursday 3) Friday 4) Tuesday

Ans: 1 (Wednesday). Day-after-tomorrow=Sun → today=Fri → yesterday=Thu. *[FLAG: today Friday → yesterday Thursday (option 2).]*

R57. (Memory-based) A is the brother of B. B is the sister of C. C is the father of D. How is A related to D?

1) Uncle 2) Father 3) Brother 4) Grandfather

Ans: 1 (Uncle). A and C are siblings (via B); C is D's father → A is D's uncle.

R58. (Memory-based) Mirror image of the number 1258 (mirror placed on the right)?

1) 8521 (reversed) 2) 1258 3) 5218 4) 2158

Ans: 1. Lateral inversion reverses order → 8521 (with digit reflection).

R59. (Memory-based) Find next: 2, 6, 12, 20, 30, ?

1) 42 2) 40 3) 44 4) 38

Ans: 1 (42). Differences 4,6,8,10,12 → 30+12 = 42. (n^2+n pattern.)

R60. (Memory-based) If South-East becomes North, North-East becomes West, then what does West become?

1) South-East 2) North-East 3) South-West 4) North-West

Ans: 2 (North-East). The compass rotates 135° anticlockwise → West maps to North-East.

R61. (Memory-based) Which number completes: 7, 14, 28, 56, ?

1) 112 2) 108 3) 96 4) 120

Ans: 1 (112). ×2 each term.

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4. Sources

Primary (verified real papers with official answer keys):

- RRB JE CBT-1 Paper — 19-02-2026 — Shift 1 (English). Local file: `RRB/Rrb/RRB-JE-CBT-1-Paper-Answer-Key---19-02-2026---S1-English.pdf`. Blocks tagged **(S1 Qn)**.
- RRB JE CBT-1 Paper — 19-02-2026 — Shift 2 (English). Local file: `RRB/Rrb/RRB-JE-CBT-1-Paper-Answer-Key---19-02-2026---S2-English-1.pdf`. Blocks tagged **(S2 Qn)**.

Internet — weightage & topic analysis:

- Oliveboard — RRB JE Subject-Wise Weightage
- PW — RRB JE Reasoning Highest-Weightage Topics
- PW — RRB JE Subject-Wise Weightage / Marks Distribution
- SSCADDA — RRB JE Mathematics Syllabus 2024
- Examolysis — RRB JE CBT-1 Exam Analysis 2024

Internet — PYQ paper repositories (for memory-based blocks R50–R61, M52–M62):

- Oliveboard — RRB JE 2025 CBT-1 Shift-wise PYPs
- Testbook — RRB JE Previous Year Papers with Answers
- CareerPower — RRB JE Previous Year Question Paper with Solutions
- Adda247 — RRB JE Previous Year Papers (shift-wise PDF)

Flag note for the book team: items marked `[FLAG]` are where my independent solution disagrees with the apparent keyed option or where OCR mangled the stem/options. Each should be reconciled against the original PDF (green-tick key) before typesetting. Memory-based items (M52–M62, R50–R61) are widely-circulated recurring patterns, not from a single verified key — treat them as practice, not certified PYQs.